



Catalytic study of the conversion of ethanol into 1,3-butadiene

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ABSTRACT

Direct synthesis of 1,3-butadiene (BD) from ethanol has been studied using magnesia–silica catalysts doped with transition metal(oxide)s. The effects of Mg/Si ratio, the synthesis procedure, and the dopant concentration were studied. It was demonstrated that modification of the magnesia–silica binary system using a consecutive impregnation step significantly increases the ethanol conversion rate and BD yield. The BD yield higher than 55 mol.% was obtained at full ethanol conversion for materials containing Cu and Ag modifiers. The influence of the reaction temperature and the ethanol concentration in the feed was also investigated. This investigation led to high BD productivity ($>0.15 \text{ g}_{\text{BD}} \text{ g}_{\text{cat}}^{-1} \text{ h}^{-1}$) and high BD concentration in the product stream ($>10,000 \text{ Vppm}$).

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1. Introduction

Today 1,3-butadiene (BD) is an important building block for the chemical industry. The polymerization of BD with itself and with other olefin monomers represents its largest commercial use. The commercially most important polymers are styrene–butadiene rubbers, polybutadiene, styrene–butadiene latex, acrylonitrile–butadiene–styrene polymer and nitrile rubber. Since the early seventies, the dominant technology for producing BD is the cracking of naphtha, where BD is obtained as a co-product. As the growth of ethylene production outpaced the growth of BD demand, the oversupply of BD led to shut down of the all on-purpose BD production technologies. The recent trends in lightening of the feedstock of steam crackers reduce BD supply and may potentially lead to a shortage in BD production in coming years.

Currently a lot of attention is paid to the usage of alternative and renewable resources for fuels and chemicals. Due to political incitements as well as emerging tax policies, there is a huge incentive to use monomers from biological origin. Ethanol is one of the most relevant potential sources of “bio”-carbon today [1]. Production of bioethanol occurs via fermentation and is industrially applied with sugar and corn feedstocks, while the second generation bioethanol from lignocellulosic feedstock is under extensive development [2]. Catalytic conversion of ethanol into BD is an old industrially proven route. From the 1920s to the early 1960s,

ethanol was converted to BD, hydrogen, and water at 400–450 °C according to a one-step process developed by Sergey Lebedev using a variety of (mixed) oxide catalysts [3–5]. High yields with selectivity to BD up to 72 mol.% have been reported [6]. Unfortunately, these reports did not disclose details concerning the composition of the catalyst. Simultaneously, Carbide and Carbon Chemicals Corporation performed the industrial manufacturing of BD from ethanol in the USA using a two-step process over 2% of tantalum oxide on silica [7]. This technology, which is also known as Ostromyslenky's process, included the partial ethanol dehydrogenation to acetaldehyde, followed by transformation of the mixture of ethanol and acetaldehyde into BD with 30–35% yield with selectivity 60%.

The mechanism of the ethanol to BD transformation is extremely complicated and is still subject of debate. However, one generally accepts the involvement of the following five principal steps (Fig. 1) [6,8–12]: (1) acetaldehyde formation from ethanol; (2) aldol condensation of acetaldehyde to acetaldol; (3) dehydration of acetaldol to crotonaldehyde; (4) Meerwein–Ponndorf–Verley reaction between crotonaldehyde and ethanol to obtain crotyl alcohol and acetaldehyde; and (5) dehydration of crotyl alcohol to BD. A possible reaction scheme of the main molecular transformations is summarized in Fig. 1 [9]. Obviously, such complex reaction network of different reaction types (e.g., dehydration, MPV reduction and aldol condensation) entails a multifunctional heterogeneous catalyst with a subtle balance of the amount and strength, and the ideal proximity of acid/base and redox sites, in order to provide the ideal kinetics.

Various materials have been proposed as catalysts to perform this reaction [9–11,13–31]. Among all catalysts that were

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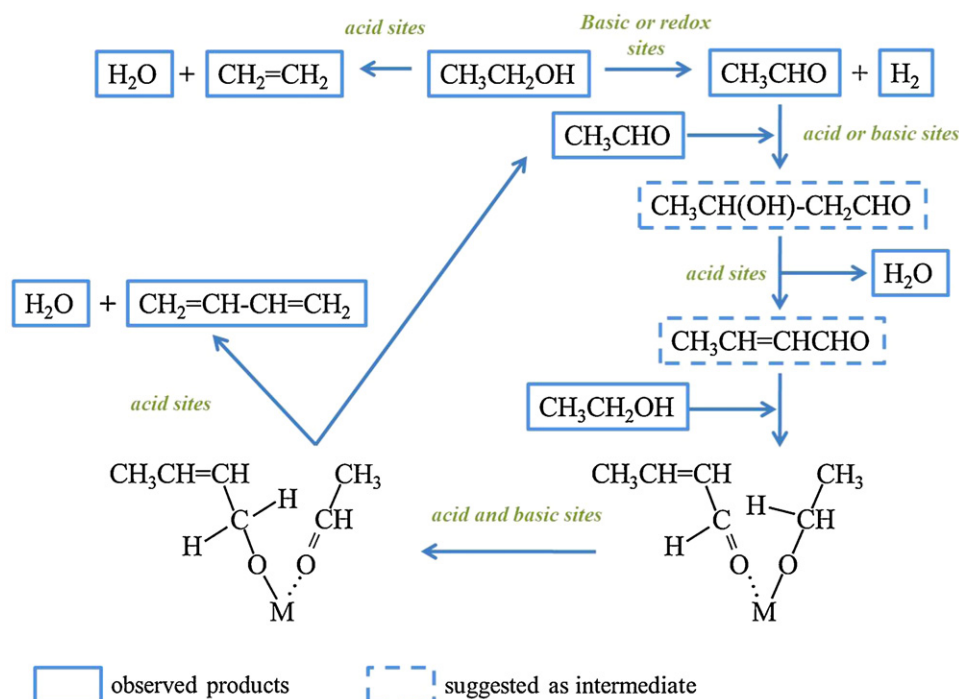


Fig. 1. Scheme of EtOH transformation [9].

suggested to perform this reaction, basic materials based on mixtures of MgO and SiO₂ were most widely studied. Several authors noticed that a Mg/Si ratio equal or higher than unity was essential to achieve high BD activity. BD yields on this type of catalyst were in the range of 9–42% with a selectivity between 30 and 84% [9,11,21,28–31]. The catalyst performance depends on both the catalyst composition and its method of preparation. The operation conditions obviously also had a serious impact on the catalytic results. Table 1 summarizes catalytic data in terms of BD yield (%), selectivity (%) and productivity ($g_{BD} g_{cat}^{-1} h^{-1}$) of various MgO/SiO₂ systems (entries 1–13) and includes the influence of temperature (entries 1–5), contact time (entries 6–9), ethanol feed concentration (entries 10–12), and presence of modifiers (entries 14–16).

Different dopants have been proposed to increase the catalysts activity and selectivity of the binary MgO–SiO₂ system and related clays such as sepiolite. Washing or impregnating the catalyst with NaOH to reduce acidity has been reported to increase the BD yields [21,32,33], but apparently this modification is only successful in

case of MgO–SiO₂ materials [16,34]. Among the other dopants, zinc, chromium, copper, manganese, nickel and silver were studied and their presence results in more active and selective catalysts for BD synthesis [10,14,15,17,27–33,35,36]. It was demonstrated that the addition of small amounts of zinc oxide led to higher BD yields, while suppressing the formation of ethylene. The optimal ZnO content to improve BD selectivity is however very subtle: above a certain concentration of the dopant, the formation of other by-products again decreases the BD yield [27]. A maximum BD yield of 45% has been reported over the ZnO doped MgO–SiO₂ system. Chromium oxide is another modifier often studied. A catalyst containing 2 mol.% of Cr₂O₃ on a 59%MgO–39%SiO₂ support was found to be the catalyst of choice among almost 500 samples tested in the one-step BD formation from ethanol [14]. Gruver in [15] observed an increasing BD yield on silver-exchanged aluminated sepiolite in comparison to the unmodified sample. A comparison of other several transition metal (such as Zn, Ni, Mn, Co and Cu) oxide supported on sepiolite has been made in [17], ZnO being the modifier of choice here.

Table 1
Overview of catalysts performance over magnesia–silica containing materials.

Entry	Sample	<i>T</i> (°C)	<i>C</i> _{EtOH} (ppm)	WHSV (h ^{−1})	TC (mol.%)	Yields (mol.%)	Selectivity (%)	BD productivity ($g g_{cat}^{-1} h^{-1}$)	BD (ppm)	Ref.
1	MgO/SiO ₂	400	–	0.3	36	19	52	0.03	–	[20]
2		410	–	0.3	39	21	54	0.03	–	
3		420	–	0.3	47	25	54	0.04	–	
4		430	–	0.3	63	32	49	0.05	–	
5		440	–	0.3	71	37	53	0.06	–	
6	MgO/SiO ₂ (Mg/Si = 3)	380	1×10^5	0.4	–	28	–	0.07	14,300	[11]
7		380	1×10^5	1.1	–	15	–	0.10	7600	
8		380	1×10^5	1.4	–	11	–	0.09	5700	
9		380	1×10^5	2.8	–	6	–	0.09	2850	
10	MgO/SiO ₂ (Mg/Si = 0.83)	350	3×10^3	0.03	53	16	30	0.003	220	[9]
11		–	6×10^3	0.06	32	8	25	0.003	230	
12		–	1×10^4	0.13	22	4	17	0.003	230	
13	MgO/SiO ₂ (Mg/Si = 1)	350	2×10^4	0.15	50	42	84	0.04	3360	[21]
14	2%Cr ₂ O ₃ –59%MgO–39%SiO ₂	400	–	0.42	68	38	56	0.08	–	[14]
15	ZnO–MgO–SiO ₂	<410	–	–	67	46	69	–	–	[27]
16	Ag–sepiolite	280	–	0.8	63	6	9	0.03	1800	[15]

A systematic study of the role of various transition metal oxide modifiers on the catalytic activity and BD selectivity of magnesia–silica materials has not been reported yet. The data currently presented in literature for such ternary $\text{MO}_x\text{--MgO--SiO}_2$ systems were obtained for materials prepared by different synthesis procedures using different testing conditions and often they are contradictory due to a lack of information. This leads to a misleading in interpretation and complicates the reproducibility of experiments. Therefore the aim of this study is to determine the influence of the nature of the redox dopant on the catalytic performance of the binary Mg–Si oxide material and to understand what could be the optimum catalysts composition in direct transformation of ethanol into BD.

2. Experimental

2.1. Catalysts preparation

Catalysts $\text{MgO}(x)\text{--SiO}_2\text{--a--e}$, where $x = \text{Mg/Si}$, were prepared by different techniques. (a) Samples $\text{MgO}(x)\text{--SiO}_2\text{--a}$ were prepared by mechanical mixing of calcined magnesium oxide and silica; (b) samples $\text{MgO}(x)\text{--SiO}_2\text{--b}$ were prepared by mechanical mixing of magnesium hydroxide and silica; (c) sample $\text{MgO}(1)\text{--SiO}_2\text{--c}$ was prepared by wet-kneading of magnesium hydroxide and silica for 4 h at room temperature as described elsewhere [9]; (d) sample $\text{MgO}(x)\text{--SiO}_2\text{--d}$ was prepared by excess solvent impregnation of silica support with a magnesium nitrate aqueous solution; (e) sample $\text{MgO}(2)\text{--SiO}_2\text{--e}$ was prepared by wet-kneading of magnesium oxide and silica in ethanol for 8 h at room temperature. Samples b–e were dried at 80 °C and calcined at 500 °C for 4 h. Grace silicagel 254 was used as the silica support in all cases except (c), where TEOS was used as the silica source.

Three series of $\text{MgO}(x)\text{--SiO}_2$ samples containing MO_x promoter, where $M = \text{Cr, Mn, Fe, Co, Ni, Cu, Zn, Ag}$ were prepared. The first series of samples $\text{MO}_x(y)/\text{MgO}(2)\text{--SiO}_2\text{--e}$, where $y = \text{M/Si}$ ratio were obtained by incipient wetness impregnation of $\text{MgO}(2)\text{--SiO}_2\text{--e}$ with appropriate amount of aqueous metal nitrate solutions. For all sample in the series the ratios $\text{M/Si} = 0.05$ and $\text{Mg/Si} = 2$. The second series of $\text{Ag}(y)/\text{MgO}(x)\text{--SiO}_2\text{--b}$ samples with different silver and magnesium loading were prepared by incipient wetness impregnation of $\text{MgO}(x)\text{--SiO}_2\text{--b}$ materials with appropriate amount of aqueous silver nitrate solution. The third series of $\text{ZnO}(y)/\text{MgO}(2)\text{--SiO}_2\text{--f}$ samples with different zinc loading were prepared by excess solvent impregnation of silica with mixed aqueous solution of magnesium and zinc nitrates at 50 °C for 4 h. All samples were dried at 80 °C and calcined at 500 °C for 4 h.

2.2. Catalysts testing

The catalytic conversion of ethanol to BD was carried out in a fixed bed continuous downstream flow quadri-reactor containing four parallel alloy reactor tubes. The reaction was performed at 350–400 °C and atmospheric pressure. Typically, 200 mg of catalyst, pelletized to a 0.25–0.5 mm fraction, were used. In order to avoid thermal decomposition of ethanol in the reactor the following procedure was applied: a layer of quartz wool was placed in the bottom of the reactor tube, then the reactor was filled with silicon carbide (SiC, 200–450 mesh particle size from Aldrich) until the middle part of the reactor, the catalyst bed was placed on top of the SiC bed and another portion of SiC was placed above the catalyst bed. Blank tests at reaction temperature without catalyst have been conducted. No significant level of conversion was detected during these tests (a small amount of acetaldehyde was observed due to thermal decomposition of ethanol). Ethanol was kept in an evaporator maintained at constant temperature and was introduced into

reactors with argon as the carrier gas. Products were analyzed by on-line gas chromatograph (GC) equipped with a CP-PoraPLOT Q-HT column and a methanator using an FID detector. Every 50 min, one tube of the quadri-reactor was selected via a multi-position valve for GC on-line analysis. Besides butadiene some amounts of carbon oxides, $\text{C}_1\text{--C}_8$ hydrocarbons, acetaldehyde, diethyl ether and other $\text{C}_3\text{--C}_4$ oxygenated compounds were found as the by-products. Carbon balance was calculated as total carbon amount in the analyzed products, divided by the total amount of carbon added and it was generally better than 90%. Catalyst coking has not been observed.

Yields (Y) of the products were calculated according the equation:

$$Y_i(\%) = 100 \cdot \frac{A_i}{\sum A_i}$$

Selectivity (S) toward i -product was calculated as follows:

$$S_i = 100 \cdot \frac{A_i}{\sum A_i - S_{\text{EtOH}}}$$

where A_i is peak area of product i in the reaction products.

2.3. Catalysts characterization

The specific surface areas of prepared samples were measured by nitrogen adsorption at 77 K using a Micromeritics Tristar 3000 instrument. Before measurements, all samples were degassed at 300 °C for 6 h. Powder XRD patterns were recorded on a STOE STADI P Combi diffractometer with focusing $\text{Ge}(111)$ monochromator ($\text{Cu K}\alpha_1$ radiation, $\lambda = 1.54056 \text{ \AA}$) in transmission geometry with an image plate position sensitive detector (IP PSD). Measurements were carried out between 2θ values of 15° and 50° with internal IP PSD resolution 0.03°, and a scan-rate 200 s. Chemical compositions were measured by ICP-OES analysis using a Iltima ICP-OES machine, and they were always close to the expected values based on the synthesis procedure.

3. Results and discussion

3.1. Catalysts characterization

The specific surface areas (S_g) determined by the BET method from the adsorption isotherm are presented in Table 2. Modification of the silica support with magnesia results in decreasing of the surface area of the samples. This decrease is more important for the samples prepared by excess solvent impregnation (samples $\text{MgO}(x)\text{--SiO}_2\text{--d}$ and $\text{ZnO}(y)/\text{MgO}(x)\text{--SiO}_2\text{--f}$) and may occur due to the formation of highly dispersed magnesia (and zinc oxide) species. These species occupy a large part of the internal and external surface of the silica particles. On the other hand, preparation of the samples by mixing of magnesium oxide or hydroxide with

Table 2
Selected analytical data for the catalysts.

Catalyst	S_g (m^2/g)
SiO_2	505
$\text{MgO}(1)\text{--SiO}_2\text{--b}$	335
$\text{MgO}(1)\text{--SiO}_2\text{--c}$	250
$\text{MgO}(1)\text{--SiO}_2\text{--d}$	98
$\text{MgO}(2)\text{--SiO}_2\text{--b}$	264
$\text{MgO}(2)\text{--SiO}_2\text{--e}$	264
$\text{CuO}(0.05)/\text{MgO}(2)\text{--SiO}_2\text{--e}$	151
$\text{ZnO}(0.05)/\text{MgO}(2)\text{--SiO}_2\text{--e}$	141
$\text{Ag}(0.05)/\text{MgO}(2)\text{--SiO}_2\text{--e}$	171
$\text{Ag}(0.05)/\text{MgO}(2)\text{--SiO}_2\text{--b}$	167
$\text{ZnO}(0.05)/\text{MgO}(2)\text{--SiO}_2\text{--f}$	79

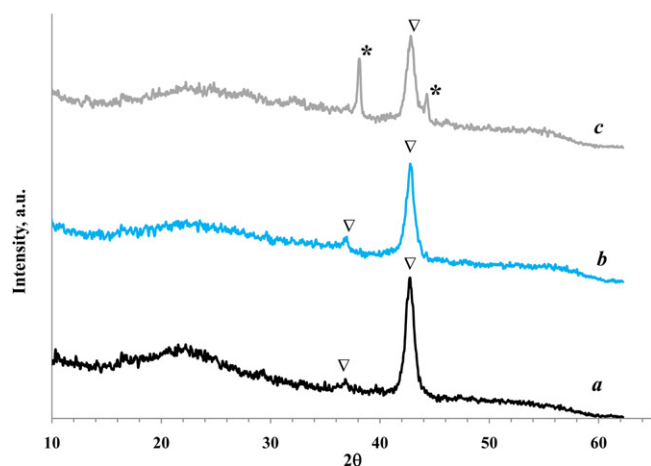


Fig. 2. XRD patterns of the prepared catalysts: a – MgO(2)–SiO₂-e; b – CuO(0.05)/MgO(2)–SiO₂-e; c – Ag(0.05)/MgO(2)–SiO₂-e; ∇ – MgO; * – Ag⁰.

silica results in formation of large magnesia particles, neighboring next to the silica particles. A decrease in specific surface area is proportional to the silica content of the sample. Further modification of the binary Mg–Si samples with transition metal oxides leads to a further decrease in specific surface area of the samples, which could be explained by the location of the oxide particles formed from nitrate salts in the silica pores.

XRD patterns of binary Mg–Si materials show the presence of crystalline MgO oxide phase with peak positions at $2\theta = 36.6^\circ$ and 42.8° , which are attributed to (1 1 1) and (2 0 0) planes of periclase (Fig. 2a). A halo in the region 2θ between 20° and 30° is due to presence of amorphous silica. The full-width at half-maximum (FWHM) of MgO reflections were slightly different for samples prepared by different synthetic procedures due to the different particle size and crystallinity. Intense and sharp XRD peaks were observed for MgO(x)–SiO₂-a samples when MgO oxide was used as a magnesium source and broad peaks were detected in case of MgO(x)–SiO₂-d samples when the MgO particles were formed from magnesium nitrate. This observation is in good agreement with the results obtained from nitrogen adsorption measurements. Increasing of the magnesia content in the binary material results in an increase of the reflection intensities. Reflections corresponding to magnesium silicates (several broad peaks with 2θ between 33° and 40°) were observed only in case of MgO(3)–SiO₂-d material with high magnesium content.

No detectable phase of transition metal oxide were observed on the diffractograms of MO_x(y)/MgO(2)–SiO₂-e samples, which

indicates relatively small or disordered oxide particles (Fig. 2b). In case of Ag(0.05)/MgO(2)–SiO₂ material containing 3.7 wt.% of silver (Fig. 2c), two additional peaks at $2\theta = 38.0^\circ$ and 44.3° appear. These peaks correspond to the (1 1 1) and (2 0 0) indexed planes of cubic Ag⁰. No peaks were detected for samples with 1 wt.% of silver with different Mg/Si ratio (Ag/Si = 0.01–0.02).

Similar to the magnesia–silica materials, several reflections attributed to magnesium silicate (or mixed zinc–magnesium silicate) were observed in the diffractograms of zinc-containing samples, which were prepared by excess solvent impregnation and ZnO(y)MgO(2)–SiO₂-f.

3.2. Catalytic activity

Typically, satisfactory mass balances (>90%) and a steady state behavior of the catalytic outcome were observed after 60 min on-stream. After this stabilization period, the ethanol conversions and products yields were stable at least during 7–8 h on-stream. It should be mentioned that some variations in ethanol conversion and products yields were observed, but only during the initial period. The same observation has been mentioned before in [9].

3.2.1. MgO–SiO₂ systems

The catalytic results of the undoped magnesia–silica materials are summarized in Fig. 3. The formation of certain amounts of ethylene and acetaldehyde (as main byproducts) was observed on bare silica gel, which is used here as the support. Silica gel is never perfectly pure and a minor impurity or isolated silanols might introduce some acidity. The activity of undoped silica in dehydration and dehydrogenation of ethanol with various selectivity toward these processes has been recognized in other work [37]. This means that the support itself is at least to a small extent responsible for the formation of some important intermediates, as well as for the formation of ethylene, being a major by-product of the ethanol to BD reaction.

The main idea behind the selected preparation procedures of the binary Mg–Si materials was to obtain samples containing both a basic (magnesia) and acidic (silica) component with different dispersion and location. Magnesia is assumed to activate the aldol condensation reaction and is also known to assist dehydrogenation, while silica catalyzes dehydration. Despite the huge difference between the preparation methods of the silica–magnesia samples, the ethanol conversion as well as the BD selectivity only differ slightly. Our experiments with the binary catalysts always showed rather low butadiene yields, viz. between 10 and 16 mol.% with BD selectivity varying between 30 and 50%, except for the MgO(1)–SiO₂-c sample, which was prepared with TEOS as silica

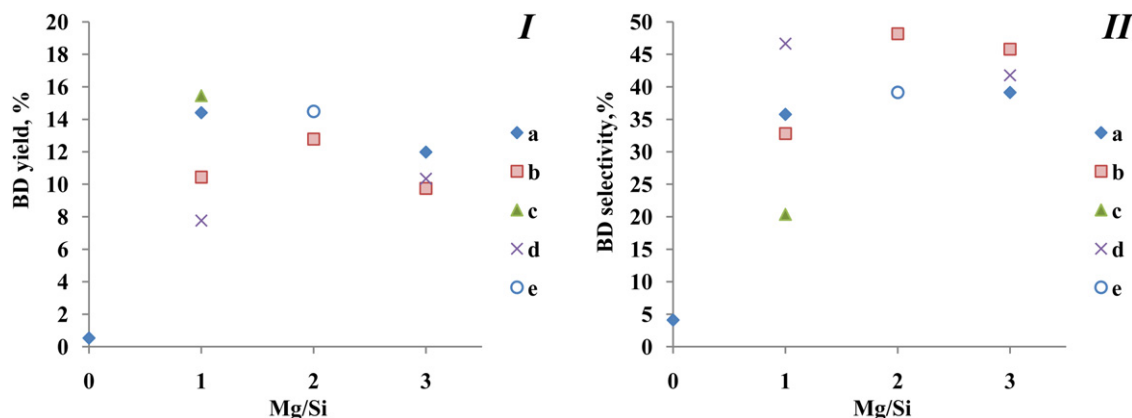


Fig. 3. Catalytic activity of the MgO–SiO₂-a–e catalysts measured at 350 °C using 1.5×10^4 ppm of EtOH in the feed with Ar flow rate of 20 ml/min: I – BD yield; II – selectivity toward BD.

source. This sample exhibited a very high ethanol conversion (up to 76 mol.% at 350 °C) with ethylene as the main reaction product (51 mol.%), but with rather poor – 20% – selectivity to BD. In addition, the amount of diethyl ether was also slightly higher in case of MgO(1)–SiO₂-c catalyst (up to 4.5%) in comparison to 2–3% yield in case of all other materials. These catalytic results evidence that even large variations in dispersion and crystallinity of the basic magnesia have little impact on the BD yield, whereas a change of the acidic component (silica) could drastically modify the BD selectivity.

Regarding the data given by prior art, the effect of magnesia–silica composition has been discussed earlier, but the data are very contradictory. For examples, Niiyama in [11] showed that catalyst containing 85 mol.% of MgO (Mg/Si = 5.7) prepared from Mg(OH)₂ and colloidal silica possess an optimal balance of acidic and basic sites. Ohnishi in [21] found that the catalytic activity of the binary material with Mg/Si = 1 is higher than that of all other compositions. In this work, when Grace silicagel 254 was used as a support, assuming both BD yield and selectivity, the optimal ratio was Mg/Si = 2.

Irrespective of the Mg/Si combinations and the synthesis procedures, magnesia–silica samples does not provide satisfactory BD yields. In other words, undoped binary Mg–Si materials could not be considered as a good candidate to convert ethanol to BD. Perhaps, additional modification with dopants could bring the solution.

3.2.2. Doped MgO–SiO₂ systems

It is well known that the addition of some amount of acetaldehyde to the ethanol feed could increase the productivity of BD [6,9,11,12,38]. Moreover, Niiyama in [11] postulated that the formation of acetaldehyde from ethanol is the rate-controlling step for this reaction. For this reason, we tried to activate the acetaldehyde formation by doping the magnesia–silica samples with various transition metal oxides. At the first step, MgO(2)–SiO₂-e material was chosen to obtain a series of samples modified with chromium, manganese, iron, cobalt, nickel, copper and zinc oxides and metallic silver. The catalytic data are given in Fig. 4 and Table 3. As expected, modification of binary Mg–Si sample with such dopant significantly increases the BD yield (entries 1–9, Table 3). All MO_x/MgO(2)–SiO₂-e catalysts except the ones doped with manganese and iron show BD yields higher than 30 mol.%. Selectivity to ethylene and diethyl ether is always lower for the doped samples.

Samples modified with manganese oxide showed the lowest activity among the tested materials (Table 3, entry 3; Fig. 4). However, selectivity toward BD is higher than 50%. In contrast, iron-containing sample exhibited a better activity in BD formation, albeit with a rather poor BD selectivity (Table 2, entry 4) due to formation of large amounts of hydrogenated C₂ and C₄ hydrocarbons, regardless of the EtOH concentration in the feed and the reaction temperature (Fig. 4II and III). The high activity in formation of hydrogenated C₄-hydrocarbons was also found for the Ni-doped samples at 400 °C (Fig. 4II and III). The same catalytic behavior has been observed for Ni-supported sepiolite in [17] compared to other transition metal dopants.

Modifying the binary support with chromium and cobalt oxide leads to a similar activity with total ethanol conversion of about 58–65% and BD yield 34–35 mol.% (Table 3, entries 2 and 5; Fig. 4). Our results for CrO_x/MgO(2)–SiO₂-e sample was lower than for the sample prepared in [14]: we obtain about 34 mol.% of butadiene (with 53% BD selectivity) at 400 °C with a 5.7×10^4 ppm of ethanol feed (Fig. 4), while the reference claims a BD yield of 38 mol.% (entry 14, Table 1). This could tentatively be explained due to the different reaction conditions of the catalytic testing and to the different preparation procedure. Regarding the results of the cobalt containing catalyst, this work did not observe the reported high activity in ethanol reforming that was mentioned in [16]. The total amount of carbon oxides and methane was below 1 mol.%

(regardless of testing conditions), and this value was comparable to that of the catalysts containing chromium and manganese and even lower than that of the others.

Among the dopants, supported CuO, ZnO and Ag catalysts exhibit the highest activity with a total ethanol conversion close to 100% and a BD yield higher than 50 mol.%, when tested at 350 °C and low ethanol feed concentration (entries 7–9 in Table 3,

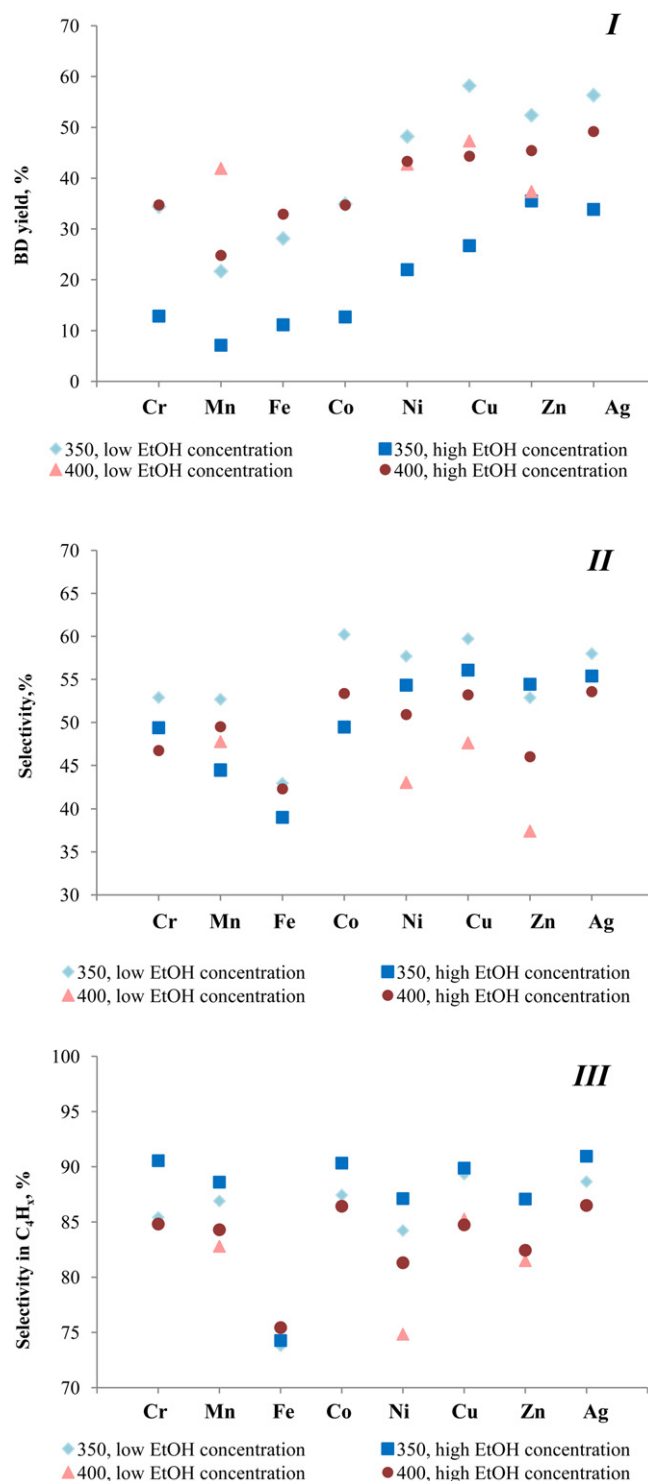


Fig. 4. Catalytic activity of the MO_x/MgO–SiO₂ catalysts measured at 350 and 400 °C using low (1.5×10^4 ppm) and high (5.7×10^4 ppm) concentration of EtOH in the feed with Ar flow rate of 20 ml/min: I – BD yield; II – selectivity toward BD; III – selectivity toward BD in C₄ hydrocarbons fraction.

Table 3

Catalytic activity of the $\text{MO}_x/\text{MgO-SiO}_2$ catalysts measured at 350 and 400 °C using low (1.5×10^4 ppm) and high (5.7×10^4 ppm) concentrations of EtOH in the feed with Ar flow rate of 20 ml/min.

Entry	M	T_{reaction} (°C)	C_{EtOH} (ppm)	TC (mol.%)	Yields (mol.%)		Selectivity (%)		BD productivity ($\text{g}_{\text{BD}} \text{g}_{\text{cat}}^{-1} \text{h}^{-1}$)	BD (ppm)
					BD	C_2H_4	BD	C_2H_4		
1	–	350	1.5×10^4	37.0	14.5	11.8	39	32	0.02	1120
2	Cr	350	1.5×10^4	65.0	34.4	11.6	53	18	0.04	2650
3	Mn	350	1.5×10^4	41.1	21.7	6.0	53	14	0.02	1670
4	Fe	350	1.5×10^4	65.5	28.1	10.4	43	16	0.03	2170
5	Co	350	1.5×10^4	58.0	34.9	7.1	60	12	0.04	2690
6	Ni	350	1.5×10^4	83.5	48.2	9.8	58	12	0.05	3720
7	Cu	350	1.5×10^4	97.5	58.2	5.5	60	6	0.07	4490
8	Zn	350	1.5×10^4	99.1	52.4	6.2	53	6	0.06	4040
9	Ag	350	1.5×10^4	97.1	56.3	5.8	58	6	0.06	4340
10	Cu	350	5.7×10^4	47.6	26.7	3.0	56	6	0.12	7630
11	Zn	350	5.7×10^4	65.2	35.5	4.2	54	6	0.15	10,140
12	Ag	350	5.7×10^4	61.1	33.9	3.1	55	5	0.14	9660
13	Cu	400	5.7×10^4	86.3	44.3	10.0	53	12	0.20	12,650
14	Zn	400	5.7×10^4	98.7	45.4	7.8	46	8	0.19	12,960
15	Ag	400	5.7×10^4	91.8	49.2	11.2	54	12	0.20	14,040

Fig. 4I). Regarding the BD selectivity, the ZnO-doped sample shows a slightly lower value due to some reforming activity of ethanol to methane and CO_x , and due to the formation of large amounts of various C_4 -oxygenated by-products. Catalytic tests performed at higher ethanol concentration in the feed showed a significant decrease in ethanol conversion and thus also in BD yield (Fig. 4I; Table 3, entries 10–12), while the BD selectivity only slightly decreases mainly due to the formation of some C_4 -oxygenated by-products. The catalysts were also tested at a higher reaction temperature, viz. 400 °C. Higher BD yields were here observed (Fig. 4I; Table 3, entries 13–15) with only a slightly decrease in BD selectivity (Fig. 4II and III).

Despite catalytic data being reported mainly as BD yields and selectivity, it is advisable also to take into account the BD concentration in the product stream (in Vppm) and the BD productivity (in $\text{g}_{\text{BD}} \text{g}_{\text{cat}}^{-1} \text{h}^{-1}$) before making a final decision on the catalytic performance. While high BD yields and selectivity are often reported at low reaction temperature and low ethanol concentration, their corresponding values for BD concentration and productivity are generally too low to be industrially relevant (see Tables 1 and 3). We suggest to work toward BD productivities higher than $0.15 \text{ g}_{\text{BD}} \text{g}_{\text{cat}}^{-1} \text{h}^{-1}$, while the concentration of BD in the product stream should be at least in the order of 10,000 Vppm or higher. Taking into account these values, Cu, Zn and Ag doped MgO(2)-SiO_2 -e are very promising materials, as they combine a high productivity and BD concentration with reasonable BD selectivity, when compared to existing catalytic systems (see Table 1).

In an attempt to further improve the catalytic data, we studied the influence of magnesia-silica support on the catalytic behavior of the Ag and Zn doped material. In order to explore if the preparation method of Mg-Si support affects the catalytic activity of such ternary $\text{MO}_x/\text{MgO-SiO}_2$ systems, another support material, viz. MgO(2)-SiO_2 -b, having lower butadiene formation activity, was used. The sample contains the same amount of silver promoter as in the previous preparation on MgO(2)-SiO_2 -e (Ag/Si=0.05).

The catalytic tests show a comparable ethanol conversion and BD selectivity for reaction temperatures and ethanol concentrations (Table 4, entries 1–2). So, despite of the activity difference between both undoped supports (Fig. 2), the addition of a small amount of Ag led to the same catalytic performance.

The effect of the silver (or zinc) promoter content and the composition of the Mg-Si support (MgO(2)-SiO_2 -b) was studied as well. Decreasing the Ag content from 3.7 wt.% (Ag/Si=0.05) to 1 wt.% (Ag/Si=0.01–0.02) did not result in a relevant change in activity and selectivity. Accordingly, the high BD yield, viz. 59 mol.%, which was obtained already for a low silver content, remained practically unchanged (Table 4, entries 2 and 4). However, variation in the Mg content results in a noticeable change in catalytic activity. For catalysts having different Mg/Si ratios, but containing the same amount (1 wt.%) of silver dopant (Table 4, entries 3–5), the total ethanol conversion and the BD selectivity changes significantly. A significant amount of unconverted acetaldehyde was observed in case of the $\text{Ag(0.01)/MgO(1)-SiO}_2$ -b sample with low Mg content: selectivity to acetaldehyde was 5%, whereas this value was lower than 2% for samples with higher Mg loading. The high acetaldehyde selectivity indicates that the total activity related to the basic sites seems insufficient to perform the subsequent aldol condensation: the more basic MgO is present in the sample, the less recoverable acetaldehyde is analyzed in the product stream. However, the BD yield reached a maximum of 59% for materials having a ratio of Mg/Si=2, while decreasing further with increasing Mg content. Probably, this is due to the lower concentration of acid sites to perform dehydration of the intermediate alcohols (Fig. 1).

Finally, few ZnO-containing catalysts were prepared by a simultaneous impregnation of silica support with both promoter and magnesium salt. The $\text{ZnO(0.05)/MgO(2)-SiO}_2$ -f sample prepared by this way showed rather poor activity (Table 4, entry 6) in comparison to $\text{ZnO(0.05)/MgO(2)-SiO}_2$ -e material prepared by consecutive modification of silica first with MgO and then with ZnO. The

Table 4

Catalytic activity of the $\text{Ag(y)/MgO(x)-SiO}_2$ and $\text{Zn(y)/MgO(2)-SiO}_2$ -f catalysts measured at 350 °C using 1.5×10^4 ppm of EtOH in the feed with Ar flow rate of 20 ml/min.

Entry	Sample	TC (mol.%)	Yields (mol.%)		Selectivity (%)		BD productivity ($\text{g}_{\text{BD}} \text{g}_{\text{cat}}^{-1} \text{h}^{-1}$)	BD (ppm)
			BD	C_2H_4	BD	C_2H_4		
1	$\text{Ag(0.05)/MgO(2)-SiO}_2$ -e	97.1	56.3	5.8	58	6	0.06	4340
2	$\text{Ag(0.05)/MgO(2)-SiO}_2$ -b	97.7	61.5	5.9	63	6	0.07	4740
3	$\text{Ag(0.01)/MgO(1)-SiO}_2$ -b	53.6	26.6	11.5	50	22	0.03	2050
4	$\text{Ag(0.01)/MgO(2)-SiO}_2$ -b	98.1	58.9	7.0	60	7	0.07	4540
5	$\text{Ag(0.02)/MgO(3)-SiO}_2$ -b	76.4	46.8	8.9	61	12	0.05	3610
6	$\text{Zn(0.05)/MgO(2)-SiO}_2$ -f	53.5	16.6	2.1	31	4	0.02	1280
7	$\text{Zn(0.1)/MgO(2)-SiO}_2$ -f	48.6	12.8	2.3	26	5	0.01	980

activity of ZnO(0.05)/MgO(2)–SiO₂-f sample was comparable to those of undoped magnesia–silica materials. It was also found that increasing of zinc oxide content does not improve catalyst efficiency (Table 4, entry 7). The reason of such catalytic behavior could be due to formation of mixed zinc–magnesium silicate. Thus the active centers of zinc oxide could not be accessible for reactants during the catalytic reaction.

4. Conclusions

The ratio between acid and basic component as well as the nature of the redox promoter has a strong influence on the catalytic performance of doped Mg–Si materials, whereas the promoter content has less impact on both activity and BD selectivity. The consecutive modification of preformed silica, first with the basic and then with the redox component, is required in order to most effectively promote the ethanol to BD reaction. Assuming parameters such as BD yield, BD selectivity, BD selectivity in the C₄-hydrocarbons fraction, BD concentration in the product stream and BD productivity, the catalysts containing silver and copper oxide as promoter were considered in this work as the best option to promote formation of BD from ethanol.

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